Maor Yehezkehely

Homework 1 – report

Query 1 shows that a very large number of companies (70,950) are associated with the officer name **"THE BEARER"**, meaning that they do not have an identifiable officer name. Such companies may be considered suspicious, and therefore further analysis of this group is worthwhile.

Query 2 reveals that these companies are not randomly distributed. Instead, some of them share the same registered address. For example, 81 companies associated with **"THE BEARER"** are registered at the address **BACK IN LINE - CONSULTING, Lisbon, Portugal**.

In Query 3, this address is examined in more detail and visualized as a graph. The graph shows that one specific company connected to this address has significantly more officers than the others.

Query 4 identifies this company as **ESPIRITO SANTO RESOURCES LIMITED**, which is associated with **76 officers**. Another interesting observation is that two additional companies among the top five also begin with the name **"ESPIRITO SANTO"**, suggesting a possible connection between these entities.

To further investigate this pattern, Query 5 examines the intermediaries associated with all companies whose names contain **"ESPIRITO SANTO"**. The results suggest that **eight of these entities are not merely separate companies that share a similar name**. Instead, they appear to be connected through a **common intermediary**, by the name "ESPIRITO SANTO SERVICES S.A. LUXEMBOURG SUCCURSALE DE LAUSANNE", that is linked to several of them.

List of Queries, tables and graphs

Query 1

```
MATCH (o:officer)-[:officer_of]-
>(e:entity)
RETURN o.name AS officer, count(e)
AS companies
ORDER BY companies DESC
LIMIT 10
```

officer	companies
THE BEARER	70950
EL PORTADOR	9326
Portcullis TrustNet (Samoa) Limited	4237
MOSSFON SUBSCRIBERS LTD.	3959
Reid Services Limited - Cayman	3003
Bearer 1	2655
Standard Directors Ltd.	2024
Execorp Limited	2000
CASCADO AG	1654
Sharecorp Limited	1649

Query 2

```
MATCH (:officer {name: "THE BEARER"})-[:officer_of]->(e:entity)-
[:registered_address]->(a:address)
RETURN a.address AS address, count(e) AS companies
ORDER BY companies DESC
LIMIT 10
```

address	companies
BACK IN LINE - CONSULTING, LDA RUA DE SAO BERNARDO N°62 1249-092 LISBOA PORTUGAL	81

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APV ADVOCATS I ECONOMISTES ASSOCIATS CARRER PAU CASALS EDIFICI CORNELLA 2 2A PLANTA, DESPATX 2 ANDORRA LA VELLA PRINCIPAT D' ANDORRA	79
DIARIO ASSETS S.A. AKARA BUILDING 24 DE CASTRO STREET WICKHAMS CAY I ROAD TOWN TORTOLA BRITISH VIRGIN ISLANDS	66
JOSE ANTONIO CORRALES ROMEO C/ DOCTOR ESQUERDO, 3 - 1º IZQUIERDA 28028 MADRID ESPAÑA **S.I.** MADRID SPAIN	64
ARAYGI & MAALOULY HORCH TABET; ST GEORGE CTR.; 5TH FLOOR P.O. BOX 55552 BEIRUT - LEBANON	61
NOMEA - NOTTER MÉGEVAND & ASSOCIÉS RUE DE LA ROSERAIE; 76A 1205; GENEVA SWITZERLAND	43
WPS SA - WEALTH PLANNING SERVICES GÉNÉRAL-DUFOUR 22 1204 GENEVE SWITZERLAND *S.I.*	40
BANQUE J. SAFRA SARASIN S.A. RUE DU RHÔNE 70 CASE POSTALE 5809 1211 GENÈVE 11, SWITZERLAND	39
DUKRE ASSET MANAGEMENT S.A. 3; RUE DE LA VALLEE 1204; GENEVA SWITZERLAND	37
PRIME PARTNERS S.A. CASE POSTALE 1987 1211 GENEVA 1 SWITZERLAND	34

Query 3

```
MATCH (o:officer {name:"THE BEARER"})-[r1:officer_of]->(e:entity)-
[r2:registered_address]->(a:address)
WHERE a.address CONTAINS "BACK IN LINE"
RETURN o, e, a, r1, r2
```

Nodes (101)

* (101) address (1) entity (19) officer (81)

Relationships (100)

* (100) officer_of (81) registered_address (19)

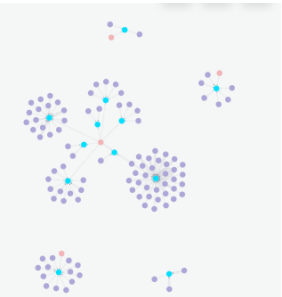
Query 4

```
MATCH (o:officer)-[:officer_of]-
>(e:entity)-[:registered_address]-
>(a:address)
WHERE a.address CONTAINS "BACK IN
LINE"
RETURN e.name AS company, count(o) AS
officers
ORDER BY officers DESC
LIMIT 5
```

company	officers
ESPIRITO SANTO RESOURCES LIMITED	76
EUROAMERICAN FINANCE CORPORATION INC.	61
ESPIRITO SANTO B.V.I. PARTICIPATION LIMITED	24
ESPIRITO SANTO AGRICULTURE AND DEVELOPMENT LIMITED	12
CONTROL DEVELOPMENTS LIMITED	6

Query 5

```
MATCH (e:entity)
WHERE e.name CONTAINS "ESPIRITO SANTO"
MATCH (o:officer)-[r1:officer_of]->(e)
OPTIONAL MATCH (i:intermediary)-[r2:intermediary_of]->(e)
RETURN e, o, i, r1, r2
```



Nodes (115)

* (115) entity (12) intermediary (4) officer (99)

Relationships (166)

* (166) intermediary_of (11) officer_of (155)

Node details

intermediary

Key	Value
<id>	4:9caf0b44-b0b6-44a8-aea3-c337db411713:1197282
name	"ESPIRITO SANTO SERVICES S.A. LUXEMBOURG SUCCURSALE DE LAUSANNE"
node_id	"11013998"
sourceID	"Panama Papers"